19.0 LICENSE RENEWAL – AGING MANAGEMENT PROGRAMS AND ACTIVITIES

19.1 INTRODUCTION

19.1.1 BACKGROUND

Renewed operating licenses for Vogtle Electric Generating Plant (VEGP) Units 1 and 2 were issued on June 3, 2009, extending the original licensed operating term by 20 years. Units 1 and 2 will enter the period of extended operation on January 17, 2027 and February 10, 2029 for Units 1 and 2, respectively.

19.1.1.1 License Renewal Rule and Process

10 CFR Part 54, the license renewal rule, establishes the procedures, criteria, and standards governing nuclear plant license renewal.

Plant systems, structures, and components (SSCs) within the scope of license renewal are defined in 10 CFR 54.4(a) as:

- Safety-related SSCs (i.e., perform a safety-related function as defined in 10 CFR 54.4(a) (1)).
- Nonsafety-related SSCs whose failure could prevent satisfactory accomplishment of safety-related functions.
- All SSCs relied on in safety analyses or plant evaluations to perform a function that demonstrates compliance with the Commission's regulations for fire protection (10 CFR 50.48), environmental qualification (10 CFR 50.49), pressurized thermal shock (10 CFR 50.61), anticipated transients without scram (10 CFR 50.62), and station blackout (10 CFR 50.63).

The license renewal rule focuses on managing the effects of aging on the passive intended functions of long-lived structures and components, and on evaluation of time-limited aging analyses (TLAA), as defined in 10 CFR 54.21. (See paragraph 19.1.1.3 for a discussion of the definition of a TLAA.)

The license renewal rule generically excludes structures and components associated only with active functions from an aging management review. Functional degradation resulting from the effects of aging on active functions is more readily determinable and detectable, and existing programs and regulatory requirements are expected to directly detect the effects of aging. The license renewal rule credits the continued applicability of existing programs and regulatory requirements, and the maintenance rule requirements (10 CFR 50.65), to monitor the performance and condition of SSCs that perform active functions.

The license renewal process includes the identification of SSCs within the scope of the license renewal rule, determining the in-scope structures and components subject to aging management review (i.e., are passive and long-lived), and assuring the effects of aging on the intended functions are adequately managed through the identification and/or development of

various aging management programs and activities. The process also includes the identification and evaluation of TLAAs, including any exemptions containing TLAAs.

The license renewal rule and the renewed operating licenses require that a summary description of the aging management programs and activities and the TLAA evaluations become part of the FSAR. To meet this requirement, sections 19.2 through 19.4 are incorporated into the FSAR. After issuance of the renewed license, 10 CFR 54.37(b) requires that, for newly identified SSCs that would have been subject to aging management review or evaluation of TLAAs in accordance with 10 CFR 54.21, the FSAR be updated to describe how the effects of aging will be managed such that the intended functions(s) in 10 CFR 54.4(b) will be effectively maintained during the period of extended operation.

19.1.1.2 Aging Management Programs

The NRC, in the Standard Review Plan for License Renewal (NUREG-1800), Appendix A.1, "Aging Management Review – Generic (Branch Technical Position RLSB-1)," describes the elements of an acceptable aging management program to the NRC staff. Additionally, NUREG-1801, "Generic Aging Lessons Learned Report," describes aging management programs that have been found acceptable to the NRC Staff to manage the aging effects of SSCs for license renewal.

In many cases, programs and activities existing at the time of the license renewal application were found adequate for managing aging for the period of extended operation. In some cases, the existing programs or activities required some degree of enhancement. Also, some new programs and activities were identified. It is important to note that only a portion of certain programs or activities may be relied upon for managing the effects of aging under the license renewal rule.

More than one program or activity may be credited to manage aging in a single system, structure, or component. Conversely, in other cases, one program or activity may manage the effects of aging in multiple systems.

19.1.1.3 Time-Limited Aging Analyses

The license renewal rule requires that TLAA be evaluated to capture certain plant-specific aging analyses explicitly based on the original 40-year operating life of the plant. In addition, the Rule requires that any exemptions based on TLAAs be identified and analyzed to justify extension of those exemptions through the renewal term.

TLAA evaluations are defined by the license renewal rule in 10 CFR 54.3 as those calculations and analyses that meet all of the following six criteria:

- Involve SSCs within the scope of license renewal.
- Consider the effects of aging.
- Involve time-limited assumptions defined by the operating term, e.g., 40 years.
- Were determined to be relevant in making a safety determination.
- Involve conclusions or provide the bases for conclusions related to the capability of the SSC to perform its intended functions, as delineated in the Rule.

- Are contained or incorporated by reference in the current licensing basis.

Once a TLAA has been identified, the Rule in 10 CFR 54.21 (c) requires it to be dispositioned by one of the following three specific criteria:

- The analyses remain valid for the period of extended operation.
- The analyses have been acceptably projected to the end of the period of extended operation.
- The effects of aging on the intended functions(s) will be adequately managed (e.g., programs or activities are in place) for the period of extended operation.

After the renewed license has been issued, 10 CFR 54.37 (b) requires that any newly identified calculations or analyses that would have been a TLAA be evaluated and a summary description placed in the FSAR.

19.1.2 AGING MANAGEMENT PROGRAMS

The following programs are credited to manage the effects of aging during the period of extended operation for license renewal and are described in section 19.2 as listed below:

- ACCW System Carbon Steel Components Program (19.2.1).
- Bolting Integrity Program (19.2.2).
- Boric Acid Corrosion Control Program (19.2.3).
- Buried Piping and Tanks Inspection Program (19.2.4).
- CASS RCS Fitting Evaluation Program (19.2.5).
- Closed Cooling Water Program (19.2.6).
- Diesel Fuel Oil Program (19.2.7).
- External Surfaces Monitoring Program (19.2.8).
- Fire Protection Program (19.2.9).
- Flow-Accelerated Corrosion Program (19.2.10).
- Flux Thimble Tube Inspection Program (19.2.11).
- Generic Letter 89-13 Program (19.2.12).
- Inservice Inspection Program (19.2.13).
- Nickel Alloy Management Program for Non-Reactor Vessel Closure Head Penetration Locations (19.2.14).

- Nickel Alloy Management Program for Reactor Vessel Closure Head Penetrations (19.2.15).
- Oil Analysis Program (19.2.16).
- One-Time Inspection Program (19.2.17).
- One-Time Inspection Program for ASME Class 1 Small Bore Piping (19.2.18).
- One-Time Inspection Program for Selective Leaching (19.2.19).
- Overhead and Refueling Crane Inspection Program (19.2.20).
- Periodic Surveillance and Preventive Maintenance Activities (19.2.21).
- Piping and Duct Internal Inspection Program (19.2.22).
- Reactor Vessel Closure Head Stud Program (19.2.23).
- Reactor Vessel Internals Program (19.2.24).
- Reactor Vessel Surveillance Program (19.2.25).
- Steam Generator Tubing Integrity Program (19.2.26).
- Steam Generator Program for Upper Internals (19.2.27).
- Water Chemistry Control Program (19.2.28).
- 10 CFR 50 Appendix J Program (19.2.29).
- Inservice Inspection Program – IWE (19.2.30).
- Inservice Inspection Program – IWL (19.2.31).
- Structural Monitoring Program (19.2.32).
- Structural Monitoring Program – Masonry Walls (19.2.33).
- Non-EQ Cables and Connections Program (19.2.34).
- Non-EQ Inaccessible Medium-Voltage Cables Program (19.2.35).
- Non-EQ Electrical Cable Connections One-Time Inspection Program (19.2.36).

19.1.3 AGING MANAGEMENT PROGRAMS – TIME LIMITED AGING ANALYSES (TLAA)

The aging management programs credited for managing the associated TLAAs during the period of extended operation are described in section 19.3 as listed below:

- Environmental Qualification Program (19.3.1).
- Fatigue Monitoring Program (19.3.2).

19.1.4 TLAA EVALUATIONS

The evaluation of TLAAs for the period of extended operation is provided in section 19.4. The TLAAs evaluated for the period of extended operation are listed below:

- Reactor Vessel Neutron Embrittlement Analyses (19.4.1).
- Metal Fatigue Analysis (19.4.2).
- Environmental Qualification Calculations (19.4.3).
- Containment Tendon Pre-Stress Analysis (19.4.4).
- Penetration Load Cycles (19.4.5).
- Other Plant Specific Analysis (19.4.6).

19.1.5 REFERENCES

1. NUREG-1800, Standard Review Plan for Review of License Renewal Applications for Nuclear Power Plants, U.S. Nuclear Regulatory Commission (Rev. 1), September 2005.
2. NUREG-1801, Generic Aging Lessons Learned (GALL) Report, U.S. Nuclear Regulatory Commission, (Rev. 1), September 2005.
3. Vogtle Electric Generating Plant Technical Specifications, Units 1 and 2.

19.2 AGING MANAGEMENT PROGRAM DESCRIPTIONS

The VEGP integrated plant assessment for license renewal identified the aging management programs credited to provide reasonable assurance that structures and components requiring an aging management review will continue to perform their intended functions consistent with the current licensing basis through the period of extended operation. This section describes the aging management programs and activities required to manage the effects of aging during the period of extended operation.

The aging management programs and activities in this section rely on the operations quality assurance program (OQAP) for VEGP and SNC for the elements of corrective action, confirmation process, and administrative controls. The VEGP quality assurance (QA) procedures, review and approval processes, and administrative controls are implemented in accordance with the requirements of 10 CFR 50, Appendix B. Corrective actions and administrative (document) control for both safety-related and nonsafety-related structures and components are accomplished per the existing corrective action program and document control program and are applicable to all aging management programs and activities that will be required during the period of extended operation. The confirmation process is part of the corrective action program and includes reviews to assure that corrective actions are adequate, tracking and reporting of corrective actions, and reviews of corrective action effectiveness. Any followup inspection required by the confirmation process is documented in accordance with the corrective action program. The corrective action, confirmation process, and administrative controls of the OQAP are applicable to all aging management programs and activities required during the period of extended operation.

19.2.1 ACCW SYSTEM CARBON STEEL COMPONENTS PROGRAM

The Auxiliary Component Cooling Water (ACCW) System Carbon Steel Components Program is a plant-specific program that manages cracking of carbon steel components exposed to ACCW through a combination of leakage monitoring and routine and periodic inspections. This includes the Units 1 and 2 ACCW systems, as well as carbon steel components serviced by these ACCW systems. The program is in response to operating experience related to nitrite-induced stress corrosion cracking (SCC) and subsequent component leakage in ACCW system components.

The program relies upon leakage detection monitoring, routine walkdowns, and periodic visual examinations. The program also includes preventive measures applicable to repairs and modifications intended to minimize crack initiation sites, lower stresses, and improve inspectability.

The ACCW System Carbon Steel Components Program will be implemented prior to the period of extended operation.

19.2.2 BOLTING INTEGRITY PROGRAM

The Bolting Integrity Program is a plant-specific program that manages cracking, loss of material, and loss of preload in mechanical bolted closures. The Bolting Integrity Program applies to safety-related and nonsafety-related bolting for pressure-retaining components within the scope of license renewal, with the exception of the reactor vessel head studs which are addressed by the Reactor Vessel Head Closure Stud Program.

Preventive aspects of the program include use of appropriate bolting and torquing practices, including control of thread lubricants. Periodic replacement of steam generator manway and handhole bolting is also included in the scope of the program as a preventive measure for managing cumulative fatigue damage for these fasteners. The program's bolting and torquing practices are based on industry guidelines, vendor recommendations, and VEGP operating experience, as appropriate for VEGP applications. Consistent with NUREG-1339 recommendations, the use of lubricants containing molybdenum disulfide, which has been specifically implicated in SCC of bolting, is prohibited by the program.

The program also includes periodic inspection of closure bolting assemblies to detect signs of leakage that may be indicative of loss of preload, loss of material, or cracking. Periodic inspection of bolted closures in conjunction with the Inservice Inspection Program and External Surfaces Monitoring Program will detect the effects of aging and joint leakage. Operator rounds and system walkdowns also identify joint leakage.

19.2.3 BORIC ACID CORROSION CONTROL PROGRAM

The Boric Acid Corrosion Control Program monitors the condition of components on which borated water may leak to ensure that borated water leakage and associated boric acid residue are identified, evaluated, and removed before any loss of intended function of affected components. The program detects boric acid leakage by periodic visual inspection of systems containing borated water for evidence of leakage and by inspection of adjacent structures and components for evidence of leakage. The program was developed in response to the recommendations of Generic Letter 88-05 and addresses operating experience contained in recent NRC generic communications.

Prior to the period of extended operation, VEGP will enhance the Boric Acid Corrosion Control Program to address the effects of borated water leakage on materials other than steels, including electrical components (e.g., electrical connectors), that are susceptible to boric acid corrosion.

19.2.4 BURIED PIPING AND TANKS INSPECTION PROGRAM

The Buried Piping and Tanks Inspection Program manages loss of material from the external surfaces of buried carbon steel, cast iron, and stainless steel components. The program includes both preventive measures and visual inspections. Preventive measures consist of coatings and wrappings which are required by design in accordance with industry standards. Buried components in the scope of license renewal will be inspected when they are excavated for maintenance or when exposed for any other reason.

Prior to entering the period of extended operation, a review will be performed to determine if at least one opportunistic or focused inspection of buried piping and tanks has been performed within the 10-year period prior to the period of extended operation. If an inspection did not occur, a focused inspection will be performed prior to the period of extended operation.

In addition, a focused inspection of buried piping and tanks will be performed within 10 years after entering the period of extended operation, unless an engineering evaluation concludes that sufficient opportunistic and focused inspections have occurred during this time to demonstrate the ability of the underground coatings to protect the underground piping and tanks from degradation.

The Buried Piping and Tanks Inspection Program will be implemented prior to the period of extended operation.

19.2.5 CASS RCS FITTING EVALUATION PROGRAM

The Cast Austenitic Stainless Steel (CASS) Reactor Coolant System (RCS) Fitting Evaluation Program manages the effects of loss of fracture toughness due to thermal aging for susceptible CASS components in the RCS. This program augments VEGP Inservice Inspection Program requirements.

This aging management program evaluates the susceptibility of CASS components to thermal aging embrittlement based on casting method, molybdenum content, and percent ferrite. Screening for susceptibility to thermal aging is not required for pump casings and valve bodies, based on the assessment documented in the letter dated May 19, 2000, from Christopher Grimes, Nuclear Regulatory Commission (NRC), to Douglas Walters, Nuclear Energy Institute (NEI), ADAMS Accession No. ML003717179. The existing ASME Section XI inspection requirements, including the alternative requirements of ASME Code Case N-481 for pump casings, are adequate for all pump casings and valve bodies.

The program provides aging management through either a flaw tolerance evaluation or enhanced volumetric examination. Additional inspection or evaluations to demonstrate that the material has adequate fracture toughness are not required for components that are not susceptible to thermal aging embrittlement.

Based on screening consistent with the process specified in NUREG-1801, Rev. 1, Section XI.M12, the VEGP components that require additional aging management under this program are the VEGP Unit 1 Loop 4 and Unit 2 Loop 1 RCP inlet elbows. For these two casings, loss of fracture toughness due to thermal aging will be managed by component-specific flaw tolerance evaluation, additional inspections, or a combination of these techniques.

The CASS RCS Fitting Evaluation Program will be implemented prior to the period of extended operation.

19.2.6 CLOSED COOLING WATER PROGRAM

The Closed Cooling Water (CCW) Program manages loss of material, cracking, and reduction of heat transfer in closed-cycle cooling water systems and the components cooled by these systems. The program is based on the EPRI CCW chemistry guidelines.

The program includes maintenance of corrosion inhibitor, pH buffering agent, and biocide concentrations. Concentrations of detrimental ionic species are monitored and reduced if necessary. Important diagnostic parameters are monitored and evaluated for significant trends. The program also uses corrosion-monitoring activities including trending of iron and copper concentrations and component inspections. Corrosion rate monitoring methods may also be used.

Prior to the period of extended operation, VEGP will enhance the CCW Program to indicate the components in each system that are most susceptible to various corrosion mechanisms and to ensure that corrosion monitoring is appropriately accomplished. This qualitative assessment will be based on an understanding of corrosion principles associated with CCW chemistries and on review of system, plant, and industry operating experience. Parameters considered in the review will include system flow parameters (focusing on identification of stagnant regions and on intermittently operated systems), normal operating temperatures, and component geometries (e.g., creviced areas).

19.2.7 DIESEL FUEL OIL PROGRAM

The Diesel Fuel Oil Program is a plant-specific program that manages loss of material in the diesel fuel oil systems for the emergency diesel generators and the diesel engine-driven fire water pumps through monitoring and maintenance of diesel fuel oil quality. The program is based on VEGP Technical Specifications requirements and supplemental requirements. Draining, cleaning, and internal condition inspections of diesel fuel oil components are implemented under other VEGP aging management programs as noted below.

- Periodic cleaning and inspection of the interior of the EDG system's diesel fuel oil storage tanks is performed under the Periodic Surveillance and Preventive Maintenance Program.

- Visual inspection of the diesel engine-driven fire water pumps fuel supply lines for leakage during diesel operation is performed under the Fire Protection Program.

- The One-Time Inspection Program describes inspections to verify the effectiveness of the Diesel Fuel Oil Program. The inspections include thickness measurements of storage tank bottom surfaces to verify that significant degradation of the tank base material is not occurring.

19.2.8 EXTERNAL SURFACES MONITORING PROGRAM

The External Surfaces Monitoring Program inspects external surfaces of mechanical system components requiring aging management for license renewal in external air environments. Surfaces constructed from materials susceptible to aging in these environments are inspected at frequencies that assure the effects of aging are managed such that system components will perform their intended function during the period of extended operation.

The program detects corrosion, flange leakage, missing or damaged insulation, damaged coatings, and indications of fretting or wear. Inspections of insulated surfaces are performed on a sampling basis, targeting areas identified by baseline inspections and operating experience as most susceptible. Accessible polymers and elastomers are also inspected.

Systems and components which are normally inaccessible and therefore not readily available for inspection are inspected when they are made accessible during outages or routine maintenance or repair or they may be inspected by remote means.

The External Surfaces Monitoring Program will be implemented prior to the period of extended operation.

SNC will perform an inspection of an emergency diesel generator fuel oil day tank vent line. This inspection will determine whether a debris screen is installed on the open end of the vent line. If a screen is installed, the inspection will further determine the material of construction of the debris screen.

19.2.9 FIRE PROTECTION PROGRAM

The Fire Protection Program includes inspections, performance testing, and condition monitoring of water- and gas-based fire protection systems, fire barriers, and fire pump diesels and their fuel oil supply components. The program manages fire protection components relied

upon for 10 CFR 50.48 compliance such that the intended functions will be maintained through the period of extended operation.

The water-based and gas-based fire suppression systems are tested and inspected in accordance with plant procedures based, in part, on the applicable National Fire Protection Association codes and standards. Periodic inspections, performance testing, and system monitoring provide an effective means to assure functionality of these components.

Diesel-driven fire pumps and fuel oil supply components are periodically inspected and tested to ensure that the diesels, pumps, and fuel oil supply components can perform their intended functions.

The fire barrier inspections include periodic visual inspection of structural fire barriers, including fire walls, floors, ceilings, fire penetration seals, and fire doors.

VEGP will implement the following enhancements to the Fire Protection Program:

- Wall thickness evaluations will be performed on water suppression piping systems using nonintrusive volumetric testing or visual inspections to ensure that wall thicknesses are within acceptable limits. Initial wall thickness evaluations will be performed before the end of the current operating term. Subsequent evaluations will be performed at plant-specific intervals during the period of extended operation. The plant-specific inspection intervals will be determined based on previous evaluations and site operating experience.

- A sample of sprinkler heads will be inspected using the guidance of NFPA 25 “Inspection, Testing and Maintenance of Water-Based Fire Protection Systems” (1998 Edition), Section 2-3.1.1 or NFPA 25 (2002 Edition), Section 5.3.1.1.1. Where sprinkler heads have been in service for 50 years, they will be replaced or representative samples from one or more sample areas will be submitted to a recognized testing laboratory for field service testing. This sampling will be performed every 10 years after the initial field service testing. The 50 years of time in service begins when the system was placed in service, not when the plant became operational.

- Prior to the period of extended operation, Fire Protection Program procedures will be revised to provide more detailed instructions for visual inspection of fire pump diesel fuel supply lines for leakage, corrosion, and general degradation while the engine is running during fire suppression system pump tests.

19.2.10 FLOW-ACCELERATED CORROSION PROGRAM

The Flow-Accelerated Corrosion (FAC) Program manages loss of material (wall thinning) due to FAC in susceptible plant piping and other components. The FAC Program is based on the guidance of NSAC-202L-R2, “Recommendations for an Effective Flow-Accelerated Corrosion Program,” including subsequent revisions. The program includes analysis to determine susceptible locations, predictive modeling techniques, baseline inspections of wall thickness, followup inspections, and repair or replacement of degraded components as necessary.

VEGP also uses the FAC Program and its inspection techniques to manage wall thinning that is occurring in piping components downstream of the steam generator blowdown demineralizers. The wall thinning has been attributed to the acidic conditions of the demineralizer effluent, not FAC.

19.2.11 FLUX THIMBLE TUBE INSPECTION PROGRAM

The Flux Thimble Tube Inspection Program manages loss of material due to fretting/wear of the incore flux detector thimble tubes. The program implements the VEGP response to NRC Bulletin No. 88-09, "Thimble Tube Thinning in Westinghouse Reactors." The program uses proven nondestructive examination techniques to monitor for wear of the flux thimble tubes.

Wear rate predictions determine the need for corrective actions such as repositioning, capping, or replacement of a flux thimble tube. The wear-rate predictions are also used to establish the interval to the next inspection.

Prior to the period of extended operation, a VEGP program procedure will be issued documenting the Flux Thimble Tube Inspection Program administration and implementing activities credited for license renewal.

19.2.12 GENERIC LETTER 89-13 PROGRAM

The Generic Letter 89-13 Program includes the activities which implement the VEGP response to the NRC recommended actions contained in Generic Letter (GL) 89-13, "Service Water System Problems Affecting Safety-Related Equipment." The Generic Letter 89-13 Program activities include mitigation, as well as performance and condition monitoring techniques, to ensure that the effects of aging on the Nuclear Service Cooling Water (NSCW) system, and on those components supplied by the NSCW system will be managed.

Prevention or mitigation of fouling and loss of material in the NSCW system and NSCW supplied components is accomplished, in part, by intermittent injection of appropriate water treatment chemicals. Other preventive and monitoring aspects of the VEGP Generic Letter 89-13 Program include periodic flushing of lines to mitigate or prevent fouling, periodic measurement of flow rates through selected components, periodic analysis of corrosion coupons, and cleaning of selected heat exchangers at regular intervals. Some components are visually inspected for fouling or loss of material. Volumetric examination may be used to detect degradation.

Prior to the period of extended operation, VEGP will implement the following enhancements to the Generic Letter 89-13 Program:

- An overall program procedure will be prepared which describes the various program activities that comprise the Generic Letter 89-13 Program and their implementing controls such as chemistry procedures, maintenance activities, scheduled surveillances, or other mechanisms.

- The VEGP Generic Letter 89-13 Program activities will include inspection of the NSCW transfer pumps' casings and bolting and NSCW cooling tower spray nozzles.

19.2.13 INSERVICE INSPECTION PROGRAM

The VEGP Inservice Inspection Program is a plant-specific program that mandates examinations, testing, and inspections of components and systems to detect deterioration and manage aging effects. The program uses periodic visual, surface, and volumetric examination and leakage tests of Class 1, 2, and 3 pressure-retaining components, their integral attachments, and supports to detect and characterize flaws.

The program is implemented in accordance with 10 CFR 50.55(a), which imposes the inservice inspection requirements of ASME Section XI for Class 1, 2, and 3 pressure-retaining components, their integral attachments, and supports. Inspection, repair, and replacement of these components are covered in Subsections IWB, IWC, IWD, and IWF, respectively.

In conformance with 10 CFR 50.55a(g)(4)(ii), and as based on ASME Inservice Inspection Program B (IWA-2432), the VEGP Inservice Inspection Program is updated at the end of each inspection interval to the latest edition and addenda of the Code specified in 10 CFR 50.55a, 12 months before the start of the inspection interval.

19.2.14 NICKEL ALLOY MANAGEMENT PROGRAM FOR NONREACTOR VESSEL CLOSURE HEAD PENETRATION LOCATIONS

The Nickel Alloy Management Program for Nonreactor Vessel Closure Head Penetration Locations is a plant-specific program that manages cracking due to primary water stress corrosion cracking (PWSCC) for nonreactor vessel head nickel alloy component locations. The overall goal of the program is to maintain plant safety and minimize the impact of PWSCC on plant availability through assessment, inspection, mitigation, and repair or replacement of susceptible components. Program development is based on MRP-126, “Generic Guidance for Alloy 600 Management.”

The program is based on the following set of implementation commitments:

1. SNC will continue to participate in industry initiatives directed at resolving PWSCC issues, such as owners’ group programs and the EPRI Materials Reliability Program.
2. SNC will comply with applicable NRC orders.
3. SNC will submit a program inspection plan for VEGP that includes implementation of applicable NRC bulletins, generic letters, and staff-accepted industry guidance. The inspection plan will be submitted to the staff for review and approval not less than 24 months prior to entering the period of extended operation for VEGP Units 1 and 2. The inspection plan will include assessments of each of the 10 aging management program elements defined in Section A.1.2.3 of NUREG-1800, Revision 1.

Nickel Alloy Management Program for Nonreactor Vessel Closure Head Penetration Locations will be fully implemented prior to the period of extended operation.

19.2.15 NICKEL ALLOY MANAGEMENT PROGRAM FOR REACTOR VESSEL CLOSURE HEAD PENETRATIONS

The Nickel Alloy Management Program for Reactor Vessel Closure Head Penetrations addresses industry concerns regarding the potential for PWSCC in nickel alloy components exposed to the reactor coolant environment. The program is based upon the requirements of NRC First Revised Order EA-03-009, which establishes requirements for susceptibility ranking and inspections. Susceptibility ranking is based on calculated effective degradation years and the results of previous inspection findings. Inspection frequencies are determined by the susceptibility category. Inspections to detect cracking include bare metal visual examinations and nonvisual techniques.

The program implements commitments for reactor vessel closure head penetrations associated with nickel alloys from NRC orders, bulletins, and generic letters and staff-accepted industry guidelines.

19.2.16 OIL ANALYSIS PROGRAM

The VEGP Oil Analysis Program ensures that the lubricating oil and hydraulic fluid environments of in-scope mechanical systems are maintained to the required quality. The Oil Analysis Program maintains lubricating oil and hydraulic fluid system contaminants (primarily water and particulates) within acceptable limits, thereby preserving an environment that is not conducive to deleterious aging effects. Program activities include sampling and analysis of lubricating oil and hydraulic fluid for detrimental contaminants.

The One-Time Inspection Program includes inspections planned to verify the effectiveness of the Oil Analysis Program.

Prior to the period of extended operation, VEGP will implement the following enhancements to the Oil Analysis Program:

- An overall program procedure or guideline will be prepared to formalize the sampling and analysis activities performed.

- Viscosity, relative level of oxidation, and flashpoint of lubricating oil samples will be determined for components where the lubricating oil is changed based on its analyzed condition (instead of being changed on a regular schedule regardless of condition). The relative level of oxidation of the lubricating oil will be monitored by analysis of the neutralization number or other appropriate parameters(s). Flashpoint monitoring will be performed for those components which have the potential for contamination of the lubricating oil with a light hydrocarbon such as fuel oil.

- When a lubricating oil sample's wear metal content screening results exceed the limits established for the wear metal content screening, the lubricating oil from that component will be subjected to additional testing. The additional testing may include detailed particle counting, elemental analysis, or analytical ferrography as necessary to validate the initial screening results and to diagnose the source of the particulates.

19.2.17 ONE-TIME INSPECTION PROGRAM

The VEGP One-Time Inspection Program provides objective evidence that an aging effect is not occurring, or that the aging effect is occurring slowly enough to not affect the component or structure intended function during the period of extended operation, and therefore will not require additional aging management.

The program uses one-time inspections of plant piping and components to verify the effectiveness of aging management programs or to confirm the insignificance of potential aging effects where:

a. An aging effect is not expected to occur but there is insufficient data to rule it out with reasonable confidence,

b. An aging effect is expected to progress very slowly in a specified environment, but localized conditions may be more adverse than specified, or

c. The characteristics of the aging effect include a long incubation period relative to the operating life of the plant.

The inspections will be performed within a window of 10 years immediately preceding the period of extended operation.

The inspections will include a baseline and a followup inspection of the effectiveness of the Boral™ neutron-absorbing panels credited in the criticality analysis for the Unit 1 spent fuel storage racks to provide reasonable assurance that the panels will continue to perform their reactivity control function during the period of extended operation. The baseline inspection will be performed within a window of 10 years immediately preceding the period of extended operation. The followup inspection will be performed at a date to be determined based on the results of the baseline inspection and relevant industry guidance, not to exceed 10 years after the baseline inspection.

19.2.18 ONE-TIME INSPECTION PROGRAM FOR ASME CLASS 1 SMALL BORE PIPING

The VEGP One-Time Inspection Program for ASME Class 1 Small Bore Piping addresses NRC concerns on the potential for cracking of Class 1 piping with a diameter less than NPS 4.

To address SCC concerns, volumetric examination of a sample population of ASME Class 1 Piping butt welds less than NPS 4 will be performed. Examination locations will be selected using a risk-based approach that will consider susceptibility, inspectability, dose, and operating experience.

To address unanticipated thermal fatigue cracking of ASME Class 1 piping less than NPS 4, VEGP will screen and evaluate pipe lines using MRP-146, “Management of Thermal Fatigue in Normally Stagnant Nonisolable Reactor Coolant System Branch Lines,” or later updated guidance. Small bore piping inspections will be performed to detect thermal fatigue only at piping locations that fail screening and are not monitored for thermal cycling.

Examinations performed by the program may be incorporated into an NRC-approved Risk-Informed Inservice Inspection Program. The inspections will be performed within a window of 10 years immediately preceding the period of extended operation.

19.2.19 ONE-TIME INSPECTION PROGRAM FOR SELECTIVE LEACHING

The VEGP One-Time Inspection Program for Selective Leaching addresses selective leaching in susceptible cast iron and copper alloy components. The program includes a one-time examination of a sample population of components most likely to exhibit selective leaching. Initial examinations will be completed prior to entering the period of extended operation. If degradation due to selective leaching is identified, additional examinations will be performed.

Examination techniques may include hardness measurement (where feasible-based on form and configuration), visual examination, metallurgical evaluation, or other proven techniques determined to be effective in identifying and assessing the extent of selective leaching.

The inspections will be performed within a window of 10 years immediately preceding the period of extended operation.

19.2.20 OVERHEAD AND REFUELING CRANE INSPECTION PROGRAM

The VEGP Overhead and Refueling Crane Inspection Program manages the effects of general corrosion and wear of the crane bridge and trolley structural girders and beams and the crane rails and support girders in the scope of license renewal.

The Overhead and Refueling Crane Inspection Program is a condition monitoring program that includes the following nuclear safety-related and quality-related material handling systems: refueling machine, fuel handling machine bridge crane, spent fuel cask bridge crane, and the containment building (reactor) polar crane.

Prior to the period of extended operation, VEGP will enhance applicable plant procedures to explicitly identify inspection of crane rails and crane structural components for loss of material due to corrosion and wear and for indication of rail misalignment.

19.2.21 PERIODIC SURVEILLANCE AND PREVENTIVE MAINTENANCE ACTIVITIES

The Periodic Surveillance and Preventive Maintenance Activities is a plant-specific program that includes existing and new periodic inspections and tests that are relied on by license renewal to manage the aging effects applicable to the components included in the program. The Periodic Surveillance and Preventive Maintenance Activities Program is generally implemented through repetitive tasks and surveillances.

Inspection and testing intervals are dependent on the component, material, and environment and take into consideration industry and plant-specific operating experience and manufacturer's recommendations.

The extent and schedule of inspections and testing assure detection of component degradation prior to loss of intended functions. Established techniques such as visual inspections are used. The following existing surveillance and maintenance activities are credited for license renewal:

- Control building control room filter unit seal inspections.
- Emergency diesel generator (EDG) diesel fuel oil storage tank cleaning and inspections.
- Steam generator blowdown trim heat exchanger inspections.
- NSCW cooling tower fill and drift eliminator testing.
- Diaphragm inspections for the boric acid storage tank, condensate storage tank, and reactor makeup water storage tank.

Prior to the period of extended operation, VEGP will enhance the Periodic Surveillance and Preventive Maintenance Activities to include the following additional surveillance and maintenance activities:

- Steam generator blowdown secondary sample bath shell inspections.
- Steam generator blowdown corrosion product monitor cooler shell inspections.
- Potable water system water heater housing inspections (for the in-scope water heaters).

19.2.22 PIPING AND DUCT INTERNAL INSPECTION PROGRAM

The VEGP Piping and Duct Internal Inspection Program manages corrosion of steel, stainless steel, and copper alloy components and degradation of elastomer components due to changes

in material properties. Inspections are normally performed concurrent with scheduled preventive maintenance, surveillance testing, and corrective maintenance activities. Specific examinations not coordinated with existing work activities may also be performed at the discretion of the program owner. Inspection locations and intervals are dependent on assessments of the likelihood of significant degradation and on current industry and plant-specific operating experiences.

Examination techniques will be appropriate to detect and assess the aging mechanism of concern and may include visual examination, nonvisual NDE such as ultrasonic testing or radiography, physical manipulation of elastomers, etc.

The Piping and Duct Internal Inspection Program will be implemented prior to the period of extended operation.

19.2.23 REACTOR VESSEL CLOSURE HEAD STUD PROGRAM

The VEGP Reactor Vessel Closure Head Stud Program provides direction for loss of material and cracking in the reactor vessel closure head studs, nuts, and washers. Program aspects include preventive measures, as described in Regulatory Guide 1.65, and condition monitoring.

Preventive measures include material controls and the use of approved lubricants. The VEGP reactor vessel head studs are fabricated from modified SA-540 Grade B24 material as specified in ASME Boiler and Pressure Vessel Code case 1605. This Code case is not specified in Regulatory Guide 1.65 but has been approved by the NRC via Regulatory Guide 1.85. VEGP actual stud material properties have ultimate tensile strengths less than 170 ksi. Reactor vessel closure head studs and nuts are lubricated with an approved, stable lubricant at each reassembly.

Condition monitoring includes examination and leakage detection consistent with the VEGP Inservice Inspection Program.

19.2.24 REACTOR VESSEL INTERNALS PROGRAM

The Reactor Vessel Internals Program is a plant-specific program that addresses material degradation issues for the VEGP reactor vessel internals.

The program will be based on the following set of implementation commitments:

a. SNC will participate in the industry program for investigating and managing aging effects on reactor vessel internals.

b. SNC will evaluate and implement the results of the industry programs, such as the EPRI Material Reliability Program (MRP), as applicable to the VEGP reactor vessel internals.

c. SNC will submit an inspection plan for the VEGP reactor vessel internals to the NRC for review and approval not less than 24 months before entering the period of extended operation for VEGP Units 1 and 2. This inspection plan will address the bases, inspection methods, and acceptance criteria associated with aging management of the reactor vessel thermal sleeves and the core support lugs (along with the associated support pads and attachment welds).

The Reactor Vessel Internals Program will be implemented prior to the period of extended operation.

19.2.25 REACTOR VESSEL SURVEILLANCE PROGRAM

The Reactor Vessel Surveillance Program manages loss of fracture toughness due to neutron embrittlement in reactor vessel alloy steel materials exposed to neutron fluence exceeding 1 x 10^{17} n/cm^2 (E > 1.0 MeV). The program is based on 10 CFR 50, Appendix H, "Reactor Vessel Material Surveillance Requirements," and ASTM E 185-82, "Standard Practice for Conducting Surveillance Tests for Light-Water Cooled Nuclear Power Reactor Vessels."

Capsules are periodically removed during the course of plant operating life. Neutron embrittlement is evaluated through surveillance capsule testing and evaluation, fluence calculations and benchmarking, and monitoring of effective full power years (EFPYs).

For both the VEGP Unit 1 and 2 reactor vessels, capsules with accumulated neutron fluence equivalent to 60 years of operation have already been pulled and tested. The remaining capsules (2 capsules in each unit) will be removed such that, at the time of removal, each of the remaining capsules will have accumulated neutron fluence that is not less than once, nor greater than twice, the peak end of life fluence expected for an additional 20-year license renewal term (80 years of operation).

The Reactor Vessel Surveillance Program will be enhanced as follows:

1. Prior to removal of the last surveillance capsule in each unit, program documents will be revised to require that tested and untested specimens from all capsules removed from the VEGP reactor vessels remain in storage.
2. Alternate dosimetry will be installed to monitor neutron fluence on the reactor vessel after removal of the last surveillance capsule in that unit. This enhancement will be implemented prior to removal of the last surveillance capsule in each unit.

19.2.26 STEAM GENERATOR TUBING INTEGRITY PROGRAM

The Steam Generator Tubing Integrity Program is a subprogram of the Steam Generator Program, which is an integrated program for managing the condition of the VEGP steam generators. The program focuses on steam generator tube integrity, tube planning, and the management and repair of steam generator tubing. The Steam Generator Program is in compliance with the program described in NEI 97-06, Steam Generator Program Guidelines, and VEGP Technical Specifications, subsection 5.5.9. Program deviations from NEI 97-06 are prepared and approved in accordance with NEI 97-06 and EPRI steam generator management program guidance.

The program includes a balance of prevention, inspection, evaluation and repair, and leakage monitoring. Major program elements include degradation assessments, inspection, integrity assessments, leakage monitoring, and chemistry controls.

19.2.27 STEAM GENERATOR PROGRAM FOR UPPER INTERNALS

The Steam Generator Program for Upper Internals is a plant-specific subprogram of the VEGP Steam Generator Program, which is an integrated program for managing the condition of the steam generators. The Steam Generator Program is in compliance with the program described in NEI 97-06, Steam Generator Program Guidelines.

The Steam Generator Program for Upper Internals includes VEGP Steam Generator Program activities associated with aging management of the steam generator upper internals components determined to be within the scope of license renewal. The program implements

inspection activities intended to detect degradation of secondary side internals needed to maintain tubing integrity and accomplish steam generator intended functions. An assessment based upon steam generator design, potential degradation mechanisms, and related VEGP and industry operating experience is performed to establish inspection requirements for secondary side internals components. The resulting inspection requirements are incorporated into the steam generator inspection plans.

19.2.28 WATER CHEMISTRY CONTROL PROGRAM

The VEGP Water Chemistry Control Program mitigates loss of material, cracking, and reduction of heat transfer in system components and structures through the control of water chemistry. The program includes control of detrimental chemical species and the addition of chemical agents.

The VEGP Water Chemistry Control Program is based on the EPRI water chemistry guidelines for primary and secondary water chemistry control.

The One-Time Inspection Program includes inspections to verify the effectiveness of the Water Chemistry Control Program.

VEGP will monitor spent fuel pool aluminum concentrations to ensure the Boral spent fuel racks will continue to perform their intended function during the period of extended operation. If adverse trends are identified, SNC will implement corrective actions. Additionally, SNC will monitor industry experience related to Boral and will take appropriate actions if significant degradation of Boral is identified.

19.2.29 10 CFR 50 APPENDIX J PROGRAM

The 10 CFR 50 Appendix J Program monitors leakage rates through the containment pressure boundary, including penetrations and access openings. Containment leak rate tests assure that leakage through the primary containment and systems and components penetrating primary containment does not exceed the allowable leakage limits specified within the VEGP Technical Specifications. Corrective actions are taken if leakage rates exceed established administrative limits for individual penetrations or the overall containment pressure boundary.

19.2.30 INSERVICE INSPECTION PROGRAM – IWE

The VEGP Inservice Inspection Program – IWE is a plant-specific program implemented in accordance with 10 CFR 50.55(a), which imposes the inservice inspection requirements of ASME Section XI, Subsection IWE. The program manages aging effects for the containment liners and its integral attachments including connecting penetrations and parts forming the leaktight boundary. The primary inspection method for the program is periodic visual examination along with limited volumetric examinations utilizing ultrasonic thickness measurements as needed.

In conformance with 10 CFR 50.55a(g)(4)(ii) and as based on ASME Inservice Inspection Program B (IWA-2432), the VEGP Inservice Inspection Program – IWE is updated at the end of each 120-month inspection interval to the latest edition and addenda of the Code specified in 10 CFR 50.55a, 12 months before the start of the inspection interval.

Prior to the period of extended operation, the VEGP Inservice Inspection Program – IWE will be revised to provide more explicit direction to the registered professional engineer for trending and evaluating conditions identified during visual examinations.

19.2.31 INSERVICE INSPECTION PROGRAM – IWL

The VEGP Inservice Inspection Program – IWL is a plant-specific program implemented in accordance with 10 CFR 50.55(a), which imposes the inservice inspection requirements of ASME Section XI Subsection IWL for Class CC components. The program manages the reinforced concrete and unbonded post-tensioning systems of the containment structures.

In conformance with 10 CFR 50.55a(g)(4)(ii) and as based on ASME Inservice Inspection Program B (IWA-2432), the VEGP Inservice Inspection Program – IWL is updated at the end of each 120-month inspection interval to the latest edition and addenda of the Code specified in 10 CFR 50.55a, 12 months before the start of the inspection interval.

Prior to the period of extended operation, the VEGP Inservice Inspection Program – IWL will be revised to provide more explicit direction to the registered professional engineer for trending and evaluating conditions identified during concrete visual examinations.

19.2.32 STRUCTURAL MONITORING PROGRAM

The VEGP Structural Monitoring Program is based on the requirements and guidance set forth in 10 CFR 50.65, “Requirements for Monitoring the Effectiveness of Maintenance at Nuclear Power Plants,” and Regulatory Guide 1.160, Rev. 2, “Monitoring the Effectiveness of Maintenance at Nuclear Power Plants.” VEGP uses the Structural Monitoring Program to monitor the condition of structures and structural components within the scope of the Maintenance Rule, thereby providing reasonable assurance that there is no loss of structure or structural component intended function.

Prior to the period of extended operation, VEGP will implement the following enhancements to the Structural Monitoring Program:

- The scope of the Structural Monitoring Program will be expanded to include the additional structures that require monitoring for license renewal.
- The scope of inspection for structures that require monitoring for license renewal will be clarified. An area-based inspection will be performed unless a detailed inspection scope is provided.
- The Structural Monitoring Program scope for hangers and supports will be clarified.
- Program requirements will be revised to include periodic groundwater monitoring to confirm that groundwater chemistry remains nonaggressive as defined in NUREG 1801.
- Underwater inspection of the NSCW cooling tower basins, including appropriate inspection and acceptance criteria, will be added to the Structural Monitoring Program.

- Guidance will be given regarding proper documentation of condition adverse to quality and its probable causes for any CR written against a finding during Structural Monitoring Program walkdown.

- For any finding (e.g., crack, leakage, etc.) guidance will be given for data to be collected and evaluated.

- More explicit direction will be given for trending of the problems.

19.2.33 STRUCTURAL MONITORING PROGRAM – MASONRY WALLS

The Structural Monitoring Program – Masonry Walls is part of the VEGP Structural Monitoring Program that implements structures monitoring requirements as specified by 10 CFR 50.65. The Masonry Wall Program manages aging of masonry walls, and structural steel restraint systems of the masonry walls, within scope of license renewal. The program includes the concrete masonry units and restraint systems used to seal and provide radiation shielding of some access openings in the Seismic Category I structures.

The program contains inspection guidelines and lists attributes that cause aging of masonry walls, which are to be monitored during structural monitoring inspections, as well as establishes examination criteria, evaluation requirements, and acceptance criteria.

The Structural Monitoring Program – Masonry Walls will be enhanced prior to the period of extended operation to include monitoring of masonry walls in the structures which are in scope for license renewal but are not currently monitored under the program.

19.2.34 NON-EQ CABLES AND CONNECTIONS PROGRAM

The Non-EQ Cables and Connections Program will be used to maintain the function of electrical cables and connections, which are not subject to the environmental qualification requirements of 10 CFR 50.49, but are exposed to adverse localized environments caused by heat, radiation, or moisture. An adverse localized environment is an environment that is significantly more severe than the service condition for the insulated cable or connection.

A representative sample of accessible insulated cables and connections within the scope of license renewal will be visually inspected for cable and connection jacket surface anomalies such as embrittlement, discoloration, and cracking. The technical basis for the sample selections of cables and connections to be inspected is provided. The scope of this sampling program includes electrical cables and connections in adverse localized environments.

The Non-EQ Cables and Connections Program will be implemented and the first inspection completed prior to the period of extended operation.

19.2.35 NON-EQ INACCESSIBLE MEDIUM-VOLTAGE CABLES PROGRAM

The Non-EQ Inaccessible Medium-Voltage Cables Program manages the aging effects for inaccessible medium-voltage cables (cables with operating voltage from 2 kV to 35 kV) in the scope of license renewal exposed to significant moisture and voltage. The aging effect of concern is “localized damage and breakdown of insulation.” The program includes periodic inspection and removal of water accumulation in cable manholes and periodic cable testing.

Manholes which retain water and contain medium-voltage cables in the scope of license renewal are periodically inspected for water collection and the accumulated water removed, as needed. The frequency of inspection is based on actual plant-experience but at least once every 2 years.

In-scope medium-voltage cables exposed to significant moisture and voltage are tested at least once every 10 years to provide an indication of the condition of the conductor insulation. The specific test performed is a proven test for detecting deterioration of the insulation system due to wetting.

The Non-EQ Inaccessible Medium-Voltage Cables Program will be implemented and the first inspections completed prior to the period of extended operation.

19.2.36 NON-EQ ELECTRICAL CABLE CONNECTIONS ONE-TIME INSPECTION PROGRAM

The Non-EQ Cable Connections One-Time Inspection Program is a plant-specific program that performs one-time inspections on a sample of bolted connections in the scope of license renewal to confirm that loosening of electrical connections is not an aging effect requiring additional aging management during the period of extended operation. The program inspects for loosening of bolted connections due to thermal cycling, ohmic heating, electrical transients, vibration, chemical contamination, corrosion, and oxidation.

The factors considered for sample selection are application (medium and low voltage, defined as < 35 kV), circuit loading (high loading), and location (high temperature, high humidity, vibration, etc.). The technical basis for the sample selections will be documented. Inspection methods may include thermography, contact resistance testing, or appropriate methods including visual inspection based on plant configuration and industry guidance.

The inspections will be performed within a window of 10 years immediately preceding the period of extended operation.

19.2.37 REFERENCES

1. NUREG-1800, Standard Review Plan for Review of License Renewal Applications for Nuclear Power Plants, U. S. Nuclear Regulatory Commission, (Rev. 1), September 2005.
2. NUREG-1801, Generic Aging Lessons Learned (GALL) Report, U. S. Nuclear Regulatory Commission, (Rev. 1), September 2005.
3. Vogtle Electric Generating Plant Technical Specifications, Units 1 and 2.

19.3 AGING MANAGEMENT PROGRAMS – TIME LIMITED AGING ANALYSES (TLAA)

19.3.1 ENVIRONMENTAL QUALIFICATION PROGRAM

The Environmental Qualification (EQ) Program implements the requirements of 10 CFR 50.49. The EQ Program has been established to demonstrate that certain electrical components located in harsh plant environments are qualified to perform their safety functions in those harsh environments, consistent with 10 CFR 50.49 requirements. The EQ Program manages component thermal, radiation, and cyclical aging, as applicable, through the use of aging evaluations. The program requires action be taken before individual components in the scope of the program exceed their qualified life. Actions taken include replacement on a specified time interval of piece parts or complete components to maintain qualification and reanalysis.

As required by 10 CFR 50.49, EQ components not qualified for the current license term are to be refurbished, replaced, or have their qualification extended prior to reaching the aging limits established in the evaluation. Some aging evaluations for EQ components specify a qualification of at least 40 years and are considered TLAAs for license renewal. The EQ Program ensures that these EQ components are maintained within the bounds of their qualification bases.

19.3.2 FATIGUE MONITORING PROGRAM

The VEGP Fatigue Monitoring Program consists of two existing programs, which are the Fatigue and Cycle Monitoring Program and Thermal Stratification Data Collection. The Fatigue and Cycle Monitoring Program, also known as the VEGP Component or Cyclic Transient Limit Program (CCTLP), is described in subsection 5.5.5 of the Technical Specifications. This program provides controls to track the transient cycles to ensure that components are maintained within the design limit. The component cyclic or transient limits are provided in VEGP UFSAR paragraph 3.9.N.1. The Thermal Stratification Data Collection Program monitors for adverse thermal stratification and cycling resulting from isolation valve leakage in the normally stagnant nonisolable reactor coolant system (RCS) branch lines identified in the VEGP response to IEB 88-08. The VEGP Fatigue Monitoring Program uses a combination of cycle counting, cycle-based fatigue monitoring, and stress-based fatigue monitoring to monitor and track fatigue usage.

At least 2 years prior to the period of extended operation, the Fatigue Monitoring Program will be enhanced as follows:

1. Implementing documents will be revised to address the effect of the full structural weld overlays applied to the pressurizer spray and surge nozzles on the stress-based module calculation of cumulative usage factor (CUF).
2. The VEGP UFSAR will be revised to require fatigue monitoring of the accumulator/reactor heat removal (RHR) nozzles and pressurizer heater penetrations.
3. Implementing documents will be revised to reduce acceptable CUF values to account for environmental fatigue effects for those NUREG-6260 locations monitored for fatigue.

4. Implementing documents will be revised to explicitly require that the corrective actions initiated for exceeding an acceptance criterion include a review to identify and assess any additional affected reactor coolant pressure boundary locations.

5. SNC will revise the FatiguePro software to calculate a minimum projected value of 1 for any events that may potentially occur.

6. SNC will revise the FatiguePro initial CUF values for the Unit 1 and Unit 2 hot leg surge nozzles, pressurizer surge nozzles, and pressurizer heater penetrations to double the current values and recalculate the current and projected CUFs.

7. SNC will implement a fatigue management software program that uses six stress components in the stress-based fatigue calculation. The software will be appropriately benchmarked against an ASME NB-3200 fatigue analysis, and the stress-based fatigue monitoring locations will be modeled with the as-built configuration. The new software will be used to reproject 60-year CUF values for the monitored locations. When those locations were evaluated for environmental effects on fatigue, the new software will also be used to demonstrate that the environmental effects on fatigue will be adequately managed for those locations during the period of extended operation.

19.3.3 REFERENCES

1. NUREG-1800, Standard Review Plan for Review of License Renewal Applications for Nuclear Power Plants, U. S. Nuclear Regulatory Commission (Rev. 1), September 2005.

2. NUREG-1801, Generic Aging Lessons Learned (GALL) Report, U. S. Nuclear Regulatory Commission (Rev. 1), September 2005.

3. Vogtle Electric Generating Plant Technical Specifications, Units 1 and 2.

19.4 EVALUATION OF TIME LIMITED AGING ANALYSES (TLAA)

In accordance with 10 CFR 54.21(c), an application for a renewed operating license must include evaluation of TLAAs for the period of extended operation. This section summarizes the TLAAs identified for VEGP license renewal.

19.4.1 REACTOR VESSEL NEUTRON EMBRITTLEMENT ANALYSES

Analyses associated with embrittlement of reactor vessel materials due to neutron irradiation are TLAAs. The end-of-life (EOL) bases for these analyses are selected to bound the projected effective full-power years (EFPY) for an operating term of 60 years.

The following VEGP analyses are TLAAs that address the effects of neutron embrittlement on the VEGP reactor vessels:

- Neutron fluence.
- Upper-Shelf Energy (USE).
- Pressurized Thermal Shock (PTS).
- Adjusted Reference Temperature (ART).
- Pressure-Temperature (P-T) limits.

19.4.1.1 Neutron Fluence Calculation

The VEGP reactor vessel neutron fluence calculations were projected out to EOL for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(ii). The reactor vessel neutron fluences, including extended beltline materials, were calculated using a method satisfying the requirements set forth in Regulatory Guide 1.90, "Calculational and Dosimetry Methods for Determining Pressure Vessel Neutron Fluence," Revision 0 (March 2001). These projections are used in the USE, PTS, ART, and P-T analyses described in the sections that follow.

19.4.1.2 Upper-Shelf Energy (USE) Calculation

Charpy impact test upper-shelf absorbed energy (USE) of no less than 50 ft-lbs throughout the life of the reactor vessel, unless an approved analysis supports a lower value.

The VEGP analyses have been projected to the end of the period of extended operation for the reactor vessel materials (base materials and welds) with projected fluence exceeding 1×10^{17} n/cm^2 (MeV > 1.0). All Unit 1 and Unit 2 base materials and welds have a USE value at EOL of greater than 50 ft-lbs, which meets the acceptance criteria of 10 CFR 50, Appendix G. Therefore, these TLAAs have been shown to be acceptable for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(ii).

19.4.1.3 Pressurized Thermal Shock (PTS) Calculation

The requirements of 10 CFR 50.61 provide for protection against PTS events in pressurized water reactors. The screening criterion in 10 CFR 50.61 is 270°F for plates, forgings, and axial welds and 300°F for circumferential welds. According to this regulation, if the calculated RT_{PTS} for the reactor beltline materials is less than the specified screening criterion, then the vessel is acceptable with regard to the risk of vessel failure during postulated pressurized thermal shock transients.

The RT_{PTS} calculations for VEGP Units 1 and 2 have been projected to the end of the period of extended operation for all reactor vessel materials (base materials and welds) with projected fluence exceeding 1×10^{17} n/cm^2 (MeV > 1.0). All Unit 1 and Unit 2 base materials and welds meet the screening criteria contained in 10 CFR 50.61 at EOL. Therefore, these TLAAs have been shown to be acceptable for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(ii).

19.4.1.4 Adjusted Reference Temperature (ART) Calculation

The ART values are an input to the pressure-temperature (P-T) limit curves discussed in the following section. The calculations determining the ART for the critical locations of the reactor vessel meet the definition of the TLAA pursuant to the criteria of 10 CFR 54.3. These ART calculations have been projected through the end of the period of extended operation and the results demonstrate the beltline materials remain limiting, and the projected ART values permit adequate operating margins to P-T limits through the period of extended operation. Therefore, these TLAAs have been shown to be acceptable for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(ii).

19.4.1.5 Pressure-Temperature (P-T) Limits Calculation

Appendix G of 10 CFR Part 50 requires heatup and cooldown of the reactor pressure vessel be accomplished within established pressure and temperature limits. Plant-specific calculations establish these limits. The calculations utilize materials and fluence data obtained through plant-specific reactor surveillance capsule programs. The calculations for VEGP Units 1 and 2 meet the definition of a TLAA.

As described in the Pressure Temperature Limits Report (PTLR), the Reactor Vessel Surveillance Program updates the P-T limit curves considering the data gained from examination of surveillance specimens from capsules that SNC pulls. The content and update of the PTLR is in accordance with the requirements of subsection 5.6.6 of the VEGP Technical Specifications. When the operating conditions of each unit merit the use of a difference curve, the PTLR for that unit is updated to include P-T limit curves that bound the current level of neutron embrittlement (i.e., EFPY) for the unit. Therefore, this TLAA demonstration is made in accordance with 10 CFR 54.21(c)(1)(ii) and (iii).

The VEGP PTLR (for each unit) will be updated to address neutron embrittlement for the 60-year operating life prior to the unit entering the period of extended operation.

19.4.2 METAL FATIGUE ANALYSIS

The thermal fatigue analyses of the VEGP mechanical components have been identified as TLAAs.

19.4.2.1 ASME Section III, Class 1 Component Fatigue Analysis

The VEGP design incorporates the requirements of Section III Class 1 of the ASME Code, which requires a discrete analysis of the thermal, mechanical, and dynamic stress cycles on components that make up the reactor coolant pressure boundary. Although original design specifications commonly state that the transient conditions are for a 40-year design life, the fatigue analyses themselves are based on specified numbers of design transients, rather than on a specific operating life. Operating experience at VEGP and similar units has demonstrated that the analyzed numbers of design basis transients are, in general, conservative for a 40-year life. The Fatigue Monitoring Program monitors and tracks the transient cycles.

To address the additional operating term, the VEGP design transient cycles were projected through the period of extended operation. For the feedwater cycling, loss of charging flow, and loss of letdown and return to service transients, VEGP relies on cumulative usage factor (CUF) monitoring of the limiting component locations in lieu of cycle counting. Therefore, the CUFs were projected for these limiting locations in lieu of projecting their transient cycles. These limiting component locations are the steam generator main and auxiliary feedwater nozzles and the normal and alternate charging nozzles. The results of the cycles and CUF projections show that the original transient cycles were conservative and that the design fatigue analyses for Class 1 components and piping remain valid for 60 years.

In addition to the original design transients, fatigue loading transients and issues have been subsequently identified that are not part of the original fatigue analyses. For the pressurizer lower head and surge line, thermal stratification and insurge/outsurge transients are evaluated (IEB 88-11). Also, the impact of the reactor coolant system environment on the fatigue life of piping and components (GSI-190) requires specific evaluation for license renewal.

To address NRC IEB 88-11, the impact of thermal stratification on the fatigue usage in the surge line was evaluated for VEGP. The original evaluation showed that the surge line fatigue usage was acceptable for 40 years of operation, including the effects of thermal stratification due to insurge and outsurges from the pressurizer. For license renewal, stress-based fatigue monitoring is credited for managing the CUF of the surge line, including the effects of pressurizer insurge/outsurge and thermal stratification in both the pressurizer lower head and both surge line nozzles.

Generic Safety Issue (GSI) 190 addresses fatigue life of metal components and was closed by the NRC in December 1999. In the closure letter, however, the NRC concluded that licensees should address the effects of reactor coolant environment on the fatigue life of selected components as aging management programs are formulated in support of license renewal.

The effects of reactor coolant environment on component fatigue life for locations equivalent to those in Section 5.4 of NUREG/CR-6260 for the newer vintage Westinghouse plants have been evaluated for VEGP using the formulas from NUREG/CR-5704 for stainless steel components and from NUREG/CR-6583 for carbon and low-alloy steel components.

For the following locations, the application of the appropriate environmental factors to the design CUF values that were calculated based on the VEGP set of original design transients yielded acceptable results (e.g., CUF < 1.0):

- Reactor vessel shell and lower head.
- Reactor vessel inlet and outlet nozzles.

For the following locations, the application of the appropriate environmental factors to the CUF values that were calculated based on the VEGP set of original design transients yielded

unacceptable results without additional management. VEGP manages the environmentally adjusted fatigue CUF values for these locations using fatigue monitoring implemented by the Fatigue Monitoring Program:

- Surge line hot leg nozzle.
- Pressurizer heater penetrations.
- Pressurizer surge line nozzles.
- Charging nozzles.
- Safety injection nozzles.
- Accumulator / RHR nozzles.

NRC Branch Technical Position MEB 3-1 is the basis for the VEGP criteria for the postulation of high-energy line breaks (HELBs) with the exception of lines that have eliminated postulated breaks based on leak-before-break analysis. One of the criteria in MEB 3-1 for Class 1 piping is postulating pipe breaks at any intermediate locations where the CUF exceeds 0.1. The NRC staff has determined that this analysis qualifies as a TLAA.

The existing VEGP HELB analyses have been shown to remain valid for the period of extended operation as long as the Fatigue Monitoring Program maintains the CUF of the charging nozzles less than or equal to 1.0. Therefore, this TLAA has been demonstrated to be acceptable for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(i) and 10 CFR 54.21(c)(1)(iii).

Full structural weld overlays (FSWOL) have been installed on the pressurizer spray nozzles, pressurizer safety and relief nozzles, and the pressurizer surge nozzles. Fatigue crack growth analyses using ASME Code Section XI methodology were performed to demonstrate the fatigue qualification at the structural weld overlay regions. Reconciliation of the existing fatigue evaluation was performed for the limiting locations outside the FSWOL, and it was demonstrated that the pressurizer nozzles would still meet the applicable ASME Code Section III requirements. In summary, the reconciliation of the existing fatigue evaluation that was performed for the limiting locations outside the FSWOL is a TLAA that remains valid for the period of extended operation, because the cycles assumed will not be exceeded during 60 years of operation. Therefore, this TLAA has been demonstrated to be acceptable for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(i).

In conclusion, the VEGP fatigue TLAAs for ASME Class 1 components have been evaluated and shown to remain valid or are adequately managed for the period of extended operation, in accordance with the demonstration methods of 10 CFR 54.21(c)(1)(i) and 10 CFR 54.21(c)(1)(iii). The Fatigue Monitoring Program monitors and tracks transient cycles and their severity and performs CUF monitoring of selected components to ensure that Class 1 components are maintained within their fatigue design limits.

19.4.2.2 ASME Section III, Non-Class 1 Component Fatigue Analysis

The design of ASME III Code Class 2 and 3 piping systems at VEGP incorporates stress reduction factors for determining the acceptability of the piping design with respect to thermal stresses. Those in-scope components that are designed in accordance with ANSI B31.1 requirements also incorporate stress-reduction factors based upon an assumed number of

thermal expansion cycles. In general, 7000 full-temperature thermal cycles are assumed in the calculation of the thermal expansion stress, leading to a stress-reduction factor of 1.0 in the stress analyses.

SNC evaluated the validity of this assumption of 7000 full-temperature thermal cycles for 60 years of plant operation. The results of this evaluation indicate that the 7000-thermal cycle assumption remains valid and bounding for 60 years of operation. Therefore, the existing pipe stress calculations are valid for the extended period of operation in accordance with 10 CFR 54.21(c)(1)(i).

There are non-Class 1 fatigue evaluations that use a different method of analysis than the 7000 cycles described above. In general, those evaluations use the same cycles, or a subset of the cycles, used for the Class 1 piping. These analyses include the letdown heat exchangers, containment cooler cooling coils, and the main stream isolation valves. In each case, the analysis was determined to remain valid for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(i).

19.4.2.3 Reactor Coolant Pump Flywheel Fatigue

A calculation was performed for the VEGP reactor coolant pump flywheels which assumes that each pump will be subjected to 6000 start/stop cycles over a 60-year life. Current projections indicate that the 6000 start/stop cycles will remain bounding for 60 years of operation by a large margin. Therefore, fatigue of the reactor coolant pump flywheels is demonstrated in accordance with 10 CFR 54.21(c)(1)(i).

19.4.2.4 Fatigue of Reactor Vessel Supports

The Westinghouse Generic Technical Report WCAP 14422, Revision 2a, identifies fatigue of reactor vessel supports as a potential TLAA if the supports of the reactor vessel were constructed in accordance with the 1963 version of the AISC Code.

The reactor pressure vessel supports embedded within the primary shield wall are procured in accordance with ASME Code, Section III, Division 1, Subsection NF; however, since they are outside the ASME jurisdictional boundary, their design follows AISC specifications. Therefore, both the 1969 version of the AISC Code and ASME Code, Section III, Division 1, Subsection NF apply to the supports.

In the SER for WCAP 14422, the NRC has indicated that licensees must ensure that a version of the AISC Code later than 1963 was used. Since the design used the 1969 version of the AISC Code, the existing analysis is demonstrated to be valid for the extended term of operation in accordance with 10 CFR 54.21(c)(1)(i).

19.4.2.5 Fatigue of Steam Generator Secondary Manway and Handhole Bolts

Westinghouse performed a fatigue calculation for steam generator secondary manway and handhole bolts that assumed the same cycles used for Class 1 component fatigue evaluations. That calculation resulted in a qualified life for the manway bolts of only 20 years. In 1993, it was determined that after low-temperature rerate, the qualified life of the manway bolts would be reduced to 14.5 years. A new secondary side manway and handhole bolts fatigue evaluation was performed based on actual cycles to qualify the bolts for 40 years with rerating.

To ensure that the cycle limits for these bolts are not exceeded, SNC will replace both the secondary side manway bolts and the handhole bolts after 30 years of service, unless a less restrictive replacement schedule is developed and documented based on potential updated analyses initiated by the Bolting Integrity Program. SNC considers this fatigue evaluation a TLAA that is managed by the Bolting Integrity Program. Therefore, this TLAA is demonstrated in accordance with 10 CFR 54.21(c)(1)(iii).

19.4.2.6 Fatigue of Reactor Vessel Internals

A fatigue analysis of the reactor vessel internals was not required when VEGP was originally designed. However, as part of rerating, Westinghouse performed a fatigue calculation for reactor vessel internals that assumed the same cycles used for Class 1 component fatigue evaluations and resulted in CUFs less than 1.0 for all subcomponents evaluated.

VEGP evaluated this TLAA for the extended period of operation. Since the analysis utilized the same design transients as the Class 1 component evaluations, the evaluation of the ASME Class 1 piping and component design transient cycles is also applicable to the reactor vessel internals. The design cycles for the transients applicable to the reactor vessel internals are bounded by the RCS design cycles, therefore the reactor vessel internals fatigue analysis remains valid for the period of extended operation. This TLAA is demonstrated in accordance with 10 CFR 54.21(c)(1)(i).

19.4.3 ENVIRONMENTAL QUALIFICATION CALCULATIONS

The NRC has established environmental qualification (EQ) requirements in 10 CFR Part 50 Appendix A and in 10 CFR 50.49. The Environmental Qualification Program for VEGP has been established to demonstrate that certain electrical components are qualified to perform safety functions in the harsh environment following a DBA. Elements of the proof of qualification involve the original 40-year license period. Hence, the qualification reports and calculations that comprise the EQ Program meet the definition of a TLAA. Qualified lives for EQ components have already been determined, and these components are tracked to determine when they are nearing the end of their qualified lives. For those components that are nearing the end of their qualified lives, the EQ Program has provisions for the component to be re-evaluated for longer service, refurbished, requalified, or replaced. The EQ Program will be continued through the period of extended operation. Therefore, this TLAA is demonstrated in accordance with 10 CFR 54.21(c)(1)(iii).

19.4.4 CONTAINMENT TENDON PRESTRESS ANALYSIS

To meet the requirements on 10 CFR 50.55a (b)(2)(ix)(B), SNC uses an analysis to predict the amount of residual prestress in the containment tendons for VEGP. This analysis meets the definition of a TLAA. SNC extended the analysis to estimate the amount of residual prestress on the tendons after 60 years of operation. The analysis results conclude that acceptable containment tendon prestress will be retained throughout the period of extended operation. Therefore, adequate containment prestress for the period of extended operation is demonstrated in accordance with 10 CFR 54.21(c)(1)(ii).

Results from containment tendon surveillances conducted under the Inservice Inspection Program – IWL periodically update the analysis and confirm prestresses remain above the minimum required values.

19.4.5 PENETRATION LOAD CYCLES

A fatigue analysis was required for some of the VEGP containment penetrations. Those analyses qualify as TLAAs. Review of the transient assumptions for those evaluations against the transient assumptions for Class 1 component fatigue determined that none of the cycles assumed in the penetration fatigue analyses will be exceeded within the period of extended operation. Therefore, fatigue analyses for containment penetrations are acceptable without revision, and the TLAAs are demonstrated in accordance with 10 CFR 54.21(c)(1)(i).

19.4.6 OTHER PLANT-SPECIFIC ANALYSIS

19.4.6.1 Leak-Before-Break Analysis

Plant-specific leak-before-break (LBB) analyses have been performed for both VEGP units. These analyses provide the technical justification for changes to the structural design basis involving protection against the effects of postulated pipe ruptures and are identified as TLAAs since they include assumptions regarding fatigue cycles and material fracture toughness properties.

VEGP LBB analyses exist for the Units 1 and 2 reactor coolant loop piping, the pressurizer surge line, and the Unit 2 accumulator injection and the RHR branch connection lines.

The LBB analyses for the pressurizer surge line and the Unit 2 RHR branch connection line were reviewed and determined to be acceptable without revision for the period of extended operation. Therefore, these LBB analyses are demonstrated in accordance with 10 CFR 54.21(c)(1)(i).

The analyses for the primary coolant loops and the Unit 2 accumulator line have been evaluated and updated to address operation through 60 years, including reductions in cast material fracture toughness properties due to thermal aging. Therefore, these LBB analyses are demonstrated in accordance with 10 CFR 54.21(c)(1)(ii).

WCAP-10551-P, Addendum 1 performed an LBB evaluation for the Units 1 and 2 primary loop piping that explicitly addressed the PWSCC concern for the Alloy 82/182 welds in this piping. However, the NRC has not yet accepted the process used as adequately addressing their concerns. Once the NRC has accepted a process for addressing PWSCC of Alloy 82/182 welds in LBB evaluations and at least 2 years prior to the period of extended operation, SNC will verify the LBB evaluation in WCAP-10551-P, Addendum 1 meets the conditions of that process or have it reperformed using the acceptable process.

19.4.6.2 Fuel Oil Storage Tank Corrosion Allowance

The VEGP diesel fuel oil storage tanks and associated piping are not provided with cathodic protection; therefore, a liberal corrosion allowance was included. A calculation performed to evaluate the corrosion allowance included a 40-year assumption and has been determined to be a TLAA.

The calculation determined the depth of penetration for a hole of approximately 1/32 in. diameter (0.001 in^2) in the coating. The calculation was reviewed for license renewal, and it was determined that depth of penetration due to corrosion would not exceed the corrosion allowance during a 60-year operating life. Specifically, consideration of 60 years instead of

40 years in the calculation increases the depth of penetration due to corrosion from 25% to 51% of the corrosion allowance for the tanks and from 50% to 76% of the corrosion allowance for the pipes.

Therefore, demonstration is in accordance with 10 CFR 54.21(c)(1)(ii).

19.4.6.3 Steam Generator Tube, Loss of Material

VEGP UFSAR subsection 5.4.2 describes allowances for erosion and corrosion that are partially based upon a measured loss of material rate for 40 years. These allowances are used as inputs to demonstrate that stress limits established by Regulatory Guide 1.121 continue to be satisfied. Subsection 5.4.2 demonstrates that a large margin exists between the allowable tube wall degradation which satisfies Regulatory Guide 1.121 limits and the tube plug limits established by the VEGP Steam Generator Tubing Integrity Program. Increasing the expected corrosion allowance to address the period of extended operation has an insignificant effect on this margin. Further, steam generator tubing wall loss is managed by the Steam Generator Tubing Integrity Program and the requirements of Regulatory Guide 1.121 are considered within that program.

Therefore, this TLAA is managed by the Steam Generator Tubing Integrity Program and is demonstrated in accordance with 10 CFR 54.21(c)(1)(iii).

19.4.6.4 Cold Overpressure Protection System

As described in paragraph 5.2.2.10 of the VEGP UFSAR, VEGP has a cold overpressure mitigation system (COPS). A calculation has been performed to confirm that the setpoints will maintain the system pressure within the established limits when the pressure difference between the pressure transmitter and reactor midplane and maximum temperature/pressure instrument uncertainties are applied to the setpoints. This calculation meets the definition of a TLAA.

The P-T limit curves in the VEGP PTLR have been evaluated for 36 EFPY. When a revision to the PTLR is issued, the cold overpressure mitigation system setpoints will also be updated to reflect the period covered by the PTLR revision. Therefore, this cold overpressure mitigation setpoint calculation TLAA is demonstrated in accordance with 10 CFR 54.21(c)(1)(ii).

As described in the PTLR, the Reactor Vessel Surveillance Program updates the P-T limit curves considering the data gained from capsules SNC pulls, and the content and update of the PTLR is in accordance with VEGP Technical Specifications, subsection 5.6.6. The associated COPS setpoints are also updated as operational needs dictate to bound the current level of neutron embrittlement (i.e., EFPY) for the unit. Therefore, this TLAA demonstration is made in accordance with 10 CFR 54.21(c)(1)(ii) and (iii).

The VEGP PTLR (for each unit) will be updated to address neutron embrittlement for a 60-year operating life, including any changes to the COPS setpoints, prior to the unit entering the period of extended operation.

19.4.6.5 Underclad Cracking of the Reactor Pressure Vessel

There is no plant-specific evaluation of underclad cracking at VEGP, and no such cracks have been identified. Freedom from underclad cracking is ensured by special evaluation of the procedure qualification for cladding applied on low-alloy steel (SA-508, Class 2) in accordance

with Regulatory Guide 1.43. However, SNC conservatively includes underclad cracking as a TLAA. Analyses performed by Westinghouse in WCAP-15338 demonstrate that growth of underclad cracks in Westinghouse reactor pressure vessels (RPVs) does not represent a significant challenge to reactor vessel integrity for an operating term of 60 years. The assumptions used as inputs to WCAP-15338 are applicable to VEGP. The results of these analyses demonstrate that underclad cracking of reactor vessel components is not an aging effect requiring management for VEGP. TLAA disposition is in accordance with 10 CFR 54.21(c)(1)(i).

19.4.7 REFERENCES

1. NUREG-1800, Standard Review Plan for Review of License Renewal Applications for Nuclear Power Plants, U. S. Nuclear Regulatory Commission (Rev. 1), September 2005.
2. NUREG-1801, Generic Aging Lessons Learned (GALL) Report, U. S. Nuclear Regulatory Commission (Rev. 1), September 2005.
3. Vogtle Electric Generating Plant Unit 1 & Unit 2 Technical Specifications.